

Contents

Lab Manual — AI & DS Department Office Mini Project	2
Table of Contents	2
1. Introduction	2
1.1 Purpose	2
1.2 Functional Overview	3
1.3 Learning Outcomes	3
2. Project Structure (Structural View)	3
2.1 High-Level Tree (source only)	3
2.2 Layered Architecture	3
3. Application Flow (Top to Bottom)	4
3.1 Cold Start Sequence	4
3.2 Navigation Map	4
3.3 User Interaction Flows	4
4. Configuration Files	5
4.1 settings.gradle.kts	5
4.2 Root build.gradle.kts	5
4.3 gradle/libs.versions.toml (excerpt)	5
4.4 app/build.gradle.kts	6
4.5 AndroidManifest.xml	7
4.6 res/navigation/nav_graph.xml	7
4.7 res/menu/menu_main.xml	8
5. Resource Files (XML — values/)	9
5.1 res/values/strings.xml	9
5.2 res/values/colors.xml	10
5.3 res/values/themes.xml	10
5.4 res/values/dimens.xml	10
6. Layout Files (XML — res/layout/)	10
6.1 activity_main.xml (shell for entire app)	10
6.2 content_main.xml (NavHost — fragment container)	11
6.3 fragment_first.xml (Dashboard)	11
6.4 fragment_second.xml (Faculty list)	12
6.5 fragment_third.xml (Lab booking)	13
6.6 fragment_announcements.xml	14
6.7 item_faculty.xml (RecyclerView row)	16
6.8 item_announcement.xml (RecyclerView row)	17
6.9 dialog_faculty.xml & dialog_announcement.xml	18
7. Java Source Files (Top-to-Bottom Flow)	19
7.1 MainActivity.java — application entry (after manifest)	19
7.2 FirstFragment.java — dashboard navigation	21
7.3 Domain models	22
7.4 Data stores (singleton — loaded when first accessed)	23
7.5 SecondFragment.java — faculty UI + CRUD	25
7.6 ThirdFragment.java — lab booking	27
7.7 AnnouncementsFragment.java	28
7.8 RecyclerView adapters (bind row XML → model fields)	30
7.9 View Binding name mapping (study aid)	32
8. Complete Lifecycle Flow (Summary)	33
8.1 Activity lifecycle (MainActivity)	33
8.2 Fragment lifecycle (typical screen)	33
8.3 RecyclerView item lifecycle	33
8.4 End-to-end “Add faculty” timeline	33
9. File Relationship Diagram	33
9.1 Mermaid — component dependencies	33
9.2 ID reference table (XML → Java)	34
10. Lab Exercises for Students	35

Exercise 1 — Trace the launch path	35
Exercise 2 — String resources	35
Exercise 3 — Navigation	35
Exercise 4 — Validation	35
Exercise 5 — Lab booking enhancement	35
Exercise 6 — RecyclerView	35
Exercise 7 — Persistence (advanced)	35
Exercise 8 — Settings menu	35
Submission checklist	35
11. Prerequisites Checklist	35
11.1 Software	35
11.2 Knowledge	36
11.3 Opening the project	36
12. Troubleshooting	36
Logcat tips	36
Appendix A — Quick reference: execution order cheat sheet	37
Appendix B — Distribution notes for instructors	37

Lab Manual — AI & DS Department Office Mini Project

Course context: Android Application Development (Mini Project)

Project name: Mini Project in a Department Office — Everyday Activities of AI & DS

Package: `com.example.miniprojectinadepartmentofficeeverydayactivitiesofaids`

Version: 1.0

Table of Contents

1. Introduction
 2. Project Structure (Structural View)
 3. Application Flow (Top to Bottom)
 4. Configuration Files
 5. Resource Files (XML — values/)
 6. Layout Files (XML — res/layout/)
 7. Java Source Files (Top-to-Bottom Flow)
 8. Complete Lifecycle Flow (Summary)
 9. File Relationship Diagram
 10. Lab Exercises for Students
 11. Prerequisites Checklist
 12. Troubleshooting
-

1. Introduction

1.1 Purpose

This Android application simulates everyday activities of an **Artificial Intelligence & Data Science (AI & DS) department office**. Students use it to practice:

- **Single-Activity, multi-Fragment** architecture
- **Navigation Component** (graph-based screen flow)
- **View Binding** (type-safe access to views)
- **RecyclerView** with custom adapters
- **AlertDialog** for CRUD-style forms
- **Singleton in-memory data stores** (faculty and announcements)
- **Material Design 3** components (Toolbar, FAB, TextInputLayout, Cards)

1.2 Functional Overview

Screen	Purpose
Dashboard (FirstFragment)	Hub with three buttons: Faculty, Lab Booking, Announcements
Faculty Directory (SecondFragment)	List faculty; add, edit, delete via dialog
Lab Booking (ThirdFragment)	Enter lab name/date/time; submit shows Toast
Announcements (AnnouncementsFragment)	List announcements; add, edit, delete via dialog

Data for faculty and announcements lives in memory (FacultyDataStore, AnnouncementDataStore) and resets when the app process is killed.

1.3 Learning Outcomes

After completing this lab, you should be able to:

1. Trace execution from `AndroidManifest.xml` → `MainActivity` → `NavHostFragment` → destination fragment.
2. Relate XML layout IDs to generated binding classes and Java `findViewById` / binding usage.
3. Explain how Navigation actions connect fragments.
4. Implement list UIs with `RecyclerView.Adapter` and row layouts.
5. Use interfaces (`OnFacultyActionListener`, `OnAnnouncementActionListener`) for fragment adapter communication.

2. Project Structure (Structural View)

2.1 High-Level Tree (source only)

```
MiniProjectinaDepartmentOfficeEverydayActivitiesOfAIDS/
  build.gradle.kts          # Root Gradle plugins
  settings.gradle.kts       # Project name, modules
  gradle/libs.versions.toml  # Dependency versions catalog
  LAB_MANUAL.md             # This document
  app/
    build.gradle.kts        # App module config (SDK, viewBinding)
    src/main/
      AndroidManifest.xml   # App entry, MainActivity launcher
      java/.../
        MainActivity.java
        FirstFragment.java
        SecondFragment.java
        ThirdFragment.java
        AnnouncementsFragment.java
        Faculty.java / FacultyDataStore.java / FacultyAdapter.java
        Announcement.java / AnnouncementDataStore.java / AnnouncementAdapter.java
      res/
        layout/             # Activities, fragments, items, dialogs
        navigation/nav_graph.xml
        menu/menu_main.xml
        values/             # strings, colors, themes, dimens
        xml/                 # backup_rules, data_extraction_rules
```

2.2 Layered Architecture

Presentation: Activities, Fragments, Adapters, Dialogs

Navigation: nav_graph.xml + NavController

Domain models: Faculty, Announcement

Data: FacultyDataStore, AnnouncementDataStore (singleton)

3. Application Flow (Top to Bottom)

3.1 Cold Start Sequence

User taps app icon

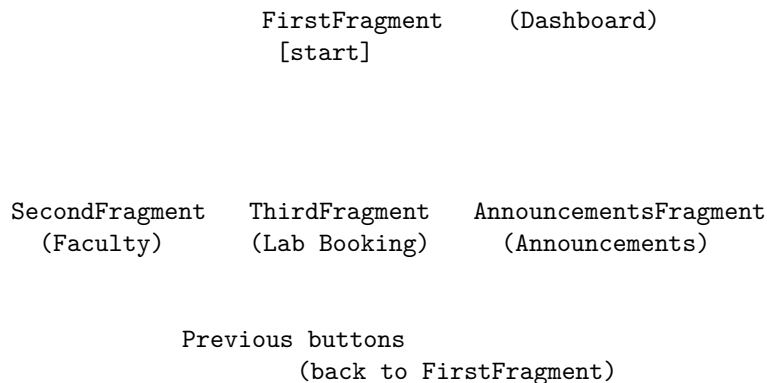
AndroidManifest: MAIN + LAUNCHER intent → MainActivity

```
MainActivity.onCreate()  
    EdgeToEdge.enable()  
    Inflate activity_main.xml (View Binding)  
    Setup Toolbar + NavController from content_main.xml  
    FAB visibility listener (visible only on Dashboard)  
    Options menu inflated (Settings)
```

NavHostFragment loads nav_graph.xml → startDestination = FirstFragment

```
FirstFragment.onCreateView() → fragment_first.xml  
FirstFragment.onViewCreated() → wire button click → navigate(...)
```

3.2 Navigation Map



3.3 User Interaction Flows

Faculty CRUD

1. Dashboard → **Faculty Directory** → SecondFragment
2. FAB + → dialog_faculty.xml in AlertDialog
3. Confirm → FacultyDataStore.addFaculty() → adapter.notifyDataSetChanged()
4. Row **Edit** / **Delete** → adapter calls SecondFragment via listener interface

Lab booking

1. Dashboard → **Lab Booking** → ThirdFragment
2. Enter lab name → **Submit Request** → Toast (validation if empty)

Announcements CRUD

1. Dashboard → **Announcements** → AnnouncementsFragment
 2. Same pattern as faculty with Announcement model and dialog_announcement.xml
-

4. Configuration Files

Configuration is read **before** any UI code runs. Order of relevance: `settings.gradle.kts` → `root build.gradle.kts` → `libs.versions.toml` → `app/build.gradle.kts` → `AndroidManifest.xml`.

4.1 settings.gradle.kts

```
pluginManagement {
    repositories {
        google {
            content {
                includeGroupByRegex("com\\.\\.android\\..*")
                includeGroupByRegex("com\\.\\.google\\..*")
                includeGroupByRegex("androidx\\..*")
            }
        }
        mavenCentral()
        gradlePluginPortal()
    }
}
plugins {
    id("org.gradle.toolchains.foojay-resolver-convention") version "1.0.0"
}
dependencyResolutionManagement {
    repositoriesMode.set(RepositoriesMode.FAIL_ON_PROJECT_REPOS)
    repositories {
        google()
        mavenCentral()
    }
}

rootProject.name = "Mini Project in a Department Office Everyday Activities Of AI & DS"
include(":app")
```

4.2 Root build.gradle.kts

```
// Top-level build file where you can add configuration options common to all sub-projects/modules.
plugins {
    alias(libs.plugins.android.application) apply false
}
```

4.3 gradle/libs.versions.toml (excerpt)

```
[versions]
agp = "9.2.1"
appcompat = "1.7.1"
material = "1.14.0"
constraintlayout = "2.2.1"
activityKtx = "1.13.0"
navigationFragment = "2.9.8"
```

```
navigationUi = "2.9.8"
```

[libraries]

```
appcompat = { group = "androidx.appcompat", name = "appcompat", version.ref = "appcompat" }  
material = { group = "com.google.android.material", name = "material", version.ref = "material" }  
constraintlayout = { group = "androidx.constraintlayout", name = "constraintlayout", version.ref = "constraintlayout" }  
activity-ktx = { group = "androidx.activity", name = "activity-ktx", version.ref = "activityKtx" }  
navigation-fragment = { group = "androidx.navigation", name = "navigation-fragment", version.ref = "navigationFragment" }  
navigation-ui = { group = "androidx.navigation", name = "navigation-ui", version.ref = "navigationUi" }
```

[plugins]

```
android-application = { id = "com.android.application", version.ref = "agp" }
```

4.4 app/build.gradle.kts

```
plugins {  
    alias(libs.plugins.android.application)  
}  
  
android {  
    namespace = "com.example.miniprojectinadepartmentofficeeverydayactivitiesofaids"  
    compileSdk {  
        version = release(36) {  
            minorApiLevel = 1  
        }  
    }  
  
    defaultConfig {  
        applicationId = "com.example.miniprojectinadepartmentofficeeverydayactivitiesofaids"  
        minSdk = 26  
        targetSdk = 36  
        versionCode = 1  
        versionName = "1.0"  
  
        testInstrumentationRunner = "androidx.test.runner.AndroidJUnitRunner"  
    }  
  
    buildTypes {  
        release {  
            isMinifyEnabled = false  
            proguardFiles(  
                getDefaultProguardFile("proguard-android-optimize.txt"),  
                "proguard-rules.pro"  
            )  
        }  
    }  
  
    compileOptions {  
        sourceCompatibility = JavaVersion.VERSION_11  
        targetCompatibility = JavaVersion.VERSION_11  
    }  
    buildFeatures {  
        viewBinding = true  
    }  
}  
  
dependencies {  
    implementation(libs.activity.ktx)
```

```

implementation(libs.appcompat)
implementation(libs.constraintlayout)
implementation(libs.material)
implementation(libs.navigation.fragment)
implementation(libs.navigation.ui)
testImplementation(libs.junit)
androidTestImplementation(libs.espresso.core)
androidTestImplementation(libs.ext.junit)
}

```

Key settings for students:

Setting	Value	Meaning
minSdk	26	Android 8.0+
viewBinding	true	Generates *Binding classes per layout
Navigation libs	2.9.8	Fragment navigation & Up button

4.5 AndroidManifest.xml

```

<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">

    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportRtl="true"
        android:theme="@style/Theme.MiniProjectInADepartmentOfficeEverydayActivitiesOfAIDS">
        <activity
            android:name=".MainActivity"
            android:exported="true"
            android:theme="@style/Theme.MiniProjectInADepartmentOfficeEverydayActivitiesOfAIDS">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />

                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
    </application>

</manifest>

```

4.6 res/navigation/nav_graph.xml

```

<?xml version="1.0" encoding="utf-8"?>
<navigation xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/nav_graph"
    app:startDestination="@id/FirstFragment">

    <fragment
        android:id="@+id/FirstFragment"

```

```

        android:name="com.example.miniprojectinadepartmentofficeeverydayactivitiesofaids.FirstFragment"
        android:label="@string/first_fragment_label"
        tools:layout="@layout/fragment_first">

        <action
            android:id="@+id/action_FirstFragment_to_SecondFragment"
            app:destination="@id/SecondFragment" />
        <action
            android:id="@+id/action_FirstFragment_to_ThirdFragment"
            app:destination="@id/ThirdFragment" />
        <action
            android:id="@+id/action_FirstFragment_to_AnnouncementsFragment"
            app:destination="@id/AnnouncementsFragment" />
    </fragment>
    <fragment
        android:id="@+id/SecondFragment"
        android:name="com.example.miniprojectinadepartmentofficeeverydayactivitiesofaids.SecondFragment"
        android:label="@string/second_fragment_label"
        tools:layout="@layout/fragment_second">

        <action
            android:id="@+id/action_SecondFragment_to_FirstFragment"
            app:destination="@id/FirstFragment" />
    </fragment>
    <fragment
        android:id="@+id/ThirdFragment"
        android:name="com.example.miniprojectinadepartmentofficeeverydayactivitiesofaids.ThirdFragment"
        android:label="@string/third_fragment_label"
        tools:layout="@layout/fragment_third">

        <action
            android:id="@+id/action_ThirdFragment_to_FirstFragment"
            app:destination="@id/FirstFragment" />
    </fragment>
    <fragment
        android:id="@+id/AnnouncementsFragment"
        android:name="com.example.miniprojectinadepartmentofficeeverydayactivitiesofaids.AnnouncementsFragment"
        android:label="Announcements"
        tools:layout="@layout/fragment_announcements">
        <action
            android:id="@+id/action_AnnouncementsFragment_to_FirstFragment"
            app:destination="@id/FirstFragment" />
    </fragment>
</navigation>

```

4.7 res/menu/menu_main.xml

```

<menu xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    tools:context="com.example.miniprojectinadepartmentofficeeverydayactivitiesofaids.MainActivity">
    <item
        android:id="@+id/action_settings"
        android:orderInCategory="100"
        android:title="@string/action_settings"
        app:showAsAction="never" />
</menu>

```

5. Resource Files (XML — values/)

Resources are compiled into R before Java runs. Strings referenced in layouts appear on screen; themes apply in onCreate before setContentView.

5.1 res/values/strings.xml

```
<resources>
    <string name="app_name">AI & DS Dept Office</string>
    <string name="action_settings">Settings</string>
    <!-- Strings used for fragments for navigation -->
    <string name="first_fragment_label">Dashboard</string>
    <string name="second_fragment_label">Faculty Directory</string>
    <string name="third_fragment_label">Lab Booking</string>
    <string name="next">Next</string>
    <string name="previous">Previous</string>

    <string name="welcome_message">Welcome to AI & DS Department Office</string>
    <string name="faculty_directory">Faculty Directory</string>
    <string name="lab_booking">Lab Booking</string>
    <string name="announcements">Announcements</string>

    <string name="faculty_list_title">Department Faculty</string>
    <string name="lab_booking_title">Book a Lab</string>
    <string name="lab_name_hint">Lab Name (e.g. AI Lab 1)</string>
    <string name="date_hint">Date</string>
    <string name="time_hint">Time</string>
    <string name="submit_booking">Submit Request</string>
    <string name="faculty_1_name">Dr. Alice Smith</string>
    <string name="faculty_1_role">Professor & Head</string>
    <string name="status_available">Status: Available</string>
    <string name="faculty_2_name">Dr. Bob Johnson</string>
    <string name="faculty_2_role">Associate Professor</string>
    <string name="status_busy">Status: In Meeting</string>

    <string name="faculty_name_hint">Faculty Name</string>
    <string name="role_hint">Role</string>
    <string name="status_hint">Status</string>
    <string name="add_faculty_title">Add Faculty</string>
    <string name="edit_faculty_title">Edit Faculty</string>
    <string name="delete_faculty_title">Confirm Deletion</string>
    <string name="delete_faculty_message">Are you sure you want to delete %1$s?</string>

    <string name="announcements_title">Latest Announcements</string>
    <string name="announcement_title_hint">Announcement Title</string>
    <string name="description_hint">Description</string>
    <string name="add_announcement_title">Add Announcement</string>
    <string name="edit_announcement_title">Edit Announcement</string>
    <string name="delete_announcement_title">Delete Announcement</string>
    <string name="delete_announcement_message">Are you sure you want to delete this announcement?</string>
    <string name="post_button">Post</string>
    <string name="confirm_button">Confirm</string>
    <string name="cancel_button">Cancel</string>
    <string name="delete_button">Delete</string>
</resources>
```

5.2 res/values/colors.xml

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
    <color name="black">#FF000000</color>
    <color name="white">#FFFFFFF</color>
</resources>
```

5.3 res/values/themes.xml

```
<resources xmlns:tools="http://schemas.android.com/tools">
    <!-- Base application theme. -->
    <style name="Base.Theme.MiniProjectInADepartmentOfficeEverydayActivitiesOfAIDS" parent="Theme.Material3
        <!-- Customize your light theme here. -->
        <!-- <item name="colorPrimary">@color/my_light_primary</item> -->
    </style>

    <style name="Theme.MiniProjectInADepartmentOfficeEverydayActivitiesOfAIDS" parent="Base.Theme.MiniProje
</resources>
```

5.4 res/values/dimens.xml

```
<resources>
    <dimen name="fab_margin">16dp</dimen>
</resources>
```

6. Layout Files (XML — res/layout/)

Layouts define the UI **before** Java attaches listeners. Execution order: inflate XML → bind views → run onCreateView logic.

6.1 activity_main.xml (shell for entire app)

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.coordinatorlayout.widget.CoordinatorLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:fitsSystemWindows="true"
    tools:context=".MainActivity">

    <com.google.android.material.appbar.AppBarLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:fitsSystemWindows="true">

        <com.google.android.material.appbar.MaterialToolbar
            android:id="@+id/toolbar"
            android:layout_width="match_parent"
            android:layout_height="?attr/actionBarSize" />

    </com.google.android.material.appbar.AppBarLayout>

    <include layout="@layout/content_main" />
```

```

        <com.google.android.material.floatingactionbutton.FloatingActionButton
            android:id="@+id/fab"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_gravity="bottom|end"
            android:layout_marginEnd="@dimen/fab_margin"
            android:layout_marginBottom="16dp"
            app:srcCompat="@android:drawable/ic_dialog_email" />

    </androidx.coordinatorlayout.widget.CoordinatorLayout>

```

6.2 content_main.xml (NavHost — fragment container)

```

<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    app:layout_behavior="@string/appbar_scrolling_view_behavior">

    <androidx.fragment.app.FragmentContainerView
        android:id="@+id/nav_host_fragment_content_main"
        android:name="androidx.navigation.fragment.NavHostFragment"
        android:layout_width="0dp"
        android:layout_height="0dp"
        app:defaultNavHost="true"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:navGraph="@navigation/nav_graph" />
</androidx.constraintlayout.widget.ConstraintLayout>

```

6.3 fragment_first.xml (Dashboard)

```

<?xml version="1.0" encoding="utf-8"?>
<androidx.core.widget.NestedScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".FirstFragment">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        android:padding="24dp"
        android:gravity="center_horizontal">

        <TextView
            android:id="@+id/textview_welcome"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="@string/welcome_message"
            android:textSize="24sp"
            android:textStyle="bold"
            android:textAlignment="center"

```

```

        android:layout_marginBottom="32dp"/>

<Button
    android:id="@+id/button_faculty"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:padding="16dp"
    android:text="@string/faculty_directory"
    android:layout_marginBottom="16dp" />

<Button
    android:id="@+id/button_lab"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:padding="16dp"
    android:text="@string/lab_booking"
    android:layout_marginBottom="16dp" />

<Button
    android:id="@+id/button_announcements"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:padding="16dp"
    android:text="@string/announcements" />

</LinearLayout>
</androidx.core.widget.NestedScrollView>

```

6.4 fragment_second.xml (Faculty list)

```

<?xml version="1.0" encoding="utf-8"?>
<androidx.coordinatorlayout.widget.CoordinatorLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".SecondFragment">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:orientation="vertical"
        android:padding="16dp">

        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="@string/faculty_list_title"
            android:textSize="20sp"
            android:textStyle="bold"
            android:layout_marginBottom="16dp"/>

        <androidx.recyclerview.widget.RecyclerView
            android:id="@+id/recycler_view_faculty"
            android:layout_width="match_parent"
            android:layout_height="0dp"
            android:layout_weight="1"

```

```

        tools:listitem="@layout/item_faculty" />

<TextView
    android:id="@+id/text_no_faculty"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_gravity="center"
    android:text="No faculty found. Click + to add."
    android:visibility="gone"
    android:layout_marginTop="32dp"/>

<Button
    android:id="@+id/button_second"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="@string/previous"
    android:layout_gravity="center_horizontal"
    android:layout_marginTop="16dp"/>

</LinearLayout>

<com.google.android.material.floatingactionbutton.FloatingActionButton
    android:id="@+id/fab_add_faculty"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_gravity="bottom|end"
    android:layout_margin="16dp"
    app:srcCompat="@android:drawable/ic_input_add"
    app:tint="@android:color/white"
    app:backgroundTint="?attr/colorPrimary"
    android:contentDescription="Add Faculty" />

</androidx.coordinatorlayout.widget.CoordinatorLayout>

```

6.5 fragment_third.xml (Lab booking)

```

<?xml version="1.0" encoding="utf-8"?>
<androidx.core.widget.NestedScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".ThirdFragment">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        android:padding="16dp">

        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="@string/lab_booking_title"
            android:textSize="20sp"
            android:textStyle="bold"
            android:layout_marginBottom="16dp"/>
    </LinearLayout>

```

```

<com.google.android.material.textfield.TextInputLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="@string/lab_name_hint"
    android:layout_marginBottom="8dp">
    <com.google.android.material.textfield.TextInputEditText
        android:id="@+id/edit_lab_name"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>
</com.google.android.material.textfield.TextInputLayout>

<com.google.android.material.textfield.TextInputLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="@string/date_hint"
    android:layout_marginBottom="8dp">
    <com.google.android.material.textfield.TextInputEditText
        android:id="@+id/edit_date"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:inputType="date"/>
</com.google.android.material.textfield.TextInputLayout>

<com.google.android.material.textfield.TextInputLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="@string/time_hint"
    android:layout_marginBottom="16dp">
    <com.google.android.material.textfield.TextInputEditText
        android:id="@+id/edit_time"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:inputType="time"/>
</com.google.android.material.textfield.TextInputLayout>

<Button
    android:id="@+id/button_submit_booking"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="@string/submit_booking" />

<Button
    android:id="@+id/button_third_back"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="@string/previous"
    android:layout_gravity="center_horizontal"
    android:layout_marginTop="16dp"/>

</LinearLayout>
</androidx.core.widget.NestedScrollView>

```

6.6 fragment_announcements.xml

```

<?xml version="1.0" encoding="utf-8"?>
<androidx.coordinatorlayout.widget.CoordinatorLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"

```

```

xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".AnnouncementsFragment">

```

```

<LinearLayout

```

```

    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">

```

```

    <TextView

```

```

        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="@string/announcements_title"
        android:textSize="20sp"
        android:textStyle="bold"
        android:layout_marginBottom="16dp"/>

```

```

    <androidx.recyclerview.widget.RecyclerView

```

```

        android:id="@+id/recycler_view_announcements"
        android:layout_width="match_parent"
        android:layout_height="0dp"
        android:layout_weight="1" />

```

```

    <TextView

```

```

        android:id="@+id/text_no_announcements"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_gravity="center"
        android:text="No announcements found. Click + to add."
        android:visibility="gone"
        android:layout_marginTop="32dp"/>

```

```

    <Button

```

```

        android:id="@+id/button_announcements_back"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="@string/previous"
        android:layout_gravity="center_horizontal"
        android:layout_marginTop="16dp"/>

```

```

</LinearLayout>

```

```

<com.google.android.material.floatingactionbutton.FloatingActionButton

```

```

    android:id="@+id/fab_add_announcement"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_gravity="bottom|end"
    android:layout_margin="16dp"
    app:srcCompat="@android:drawable/ic_input_add"
    app:tint="@android:color/white"
    app:backgroundTint="?attr/colorPrimary"
    android:contentDescription="Add Announcement" />

```

```

</androidx.coordinatorlayout.widget.CoordinatorLayout>

```


6.7 item_faculty.xml (RecyclerView row)

```
<?xml version="1.0" encoding="utf-8"?>
<com.google.android.material.card.MaterialCardView xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginBottom="8dp"
    app:cardElevation="4dp"
    app:cardCornerRadius="8dp">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal"
        android:padding="12dp"
        android:gravity="center_vertical">

        <LinearLayout
            android:layout_width="0dp"
            android:layout_height="wrap_content"
            android:layout_weight="1"
            android:orientation="vertical">

            <TextView
                android:id="@+id/text_faculty_name"
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:textSize="18sp"
                android:textStyle="bold" />

            <TextView
                android:id="@+id/text_faculty_role"
                android:layout_width="wrap_content"
                android:layout_height="wrap_content" />

            <TextView
                android:id="@+id/text_faculty_status"
                android:layout_width="wrap_content"
                android:layout_height="wrap_content" />
        </LinearLayout>

        <ImageButton
            android:id="@+id/button_edit_faculty"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:background="?attr/selectableItemBackgroundBorderless"
            android:src="@android:drawable/ic_menu_edit"
            android:padding="8dp" />

        <ImageButton
            android:id="@+id/button_delete_faculty"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:background="?attr/selectableItemBackgroundBorderless"
            android:src="@android:drawable/ic_menu_delete"
            android:padding="8dp" />

    </LinearLayout>

</com.google.android.material.card.MaterialCardView>
```



```

    </LinearLayout>
</com.google.android.material.card.MaterialCardView>

```

6.8 item_announcement.xml (RecyclerView row)

```

<?xml version="1.0" encoding="utf-8"?>
<com.google.android.material.card.MaterialCardView xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginBottom="8dp"
    app:cardElevation="2dp"
    app:cardCornerRadius="8dp">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal"
        android:padding="12dp"
        android:gravity="center_vertical">

        <LinearLayout
            android:layout_width="0dp"
            android:layout_height="wrap_content"
            android:layout_weight="1"
            android:orientation="vertical">

            <TextView
                android:id="@+id/text_announcement_title"
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:textSize="16sp"
                android:textStyle="bold" />

            <TextView
                android:id="@+id/text_announcement_description"
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:layout_marginTop="4dp" />

            <TextView
                android:id="@+id/text_announcement_date"
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:layout_marginTop="4dp"
                android:textSize="12sp"
                android:textColor="@android:color/darker_gray" />
        </LinearLayout>

        <ImageButton
            android:id="@+id/button_edit_announcement"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:background="?attr/selectableItemBackgroundBorderless"
            android:src="@android:drawable/ic_menu_edit"
            android:contentDescription="Edit Announcement"

```

```

        android:padding="8dp" />

        <ImageButton
            android:id="@+id/button_delete_announcement"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:background="?attr/selectableItemBackgroundBorderless"
            android:src="@android:drawable/ic_menu_delete"
            android:contentDescription="Delete Announcement"
            android:padding="8dp" />

    </LinearLayout>
</com.google.android.material.card.MaterialCardView>

```

6.9 dialog_faculty.xml & dialog_announcement.xml

```

<!-- dialog_faculty.xml -->
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="vertical"
    android:padding="16dp">

    <com.google.android.material.textfield.TextInputLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="@string/faculty_name_hint"
        android:layout_marginBottom="8dp">
        <com.google.android.material.textfield.TextInputEditText
            android:id="@+id/edit_faculty_name"
            android:layout_width="match_parent"
            android:layout_height="wrap_content" />
    </com.google.android.material.textfield.TextInputLayout>

    <com.google.android.material.textfield.TextInputLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="@string/role_hint"
        android:layout_marginBottom="8dp">
        <com.google.android.material.textfield.TextInputEditText
            android:id="@+id/edit_faculty_role"
            android:layout_width="match_parent"
            android:layout_height="wrap_content" />
    </com.google.android.material.textfield.TextInputLayout>

    <com.google.android.material.textfield.TextInputLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="@string/status_hint"
        android:layout_marginBottom="8dp">
        <com.google.android.material.textfield.TextInputEditText
            android:id="@+id/edit_faculty_status"
            android:layout_width="match_parent"
            android:layout_height="wrap_content" />
    </com.google.android.material.textfield.TextInputLayout>

```

```

</LinearLayout>

<!-- dialog_announcement.xml -->
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="vertical"
    android:padding="16dp">

    <com.google.android.material.textfield.TextInputLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="@string/announcement_title_hint"
        android:layout_marginBottom="8dp">
        <com.google.android.material.textfield.TextInputEditText
            android:id="@+id/edit_announcement_title"
            android:layout_width="match_parent"
            android:layout_height="wrap_content" />
    </com.google.android.material.textfield.TextInputLayout>

    <com.google.android.material.textfield.TextInputLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="@string/description_hint">
        <com.google.android.material.textfield.TextInputEditText
            android:id="@+id/edit_announcement_desc"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:inputType="textMultiLine"
            android:minLines="3"
            android:gravity="top"/>
    </com.google.android.material.textfield.TextInputLayout>

</LinearLayout>

```

7. Java Source Files (Top-to-Bottom Flow)

Java runs **after** the framework inflates XML. Order below matches runtime dependency: entry activity → navigation hub fragment → feature fragments → models → stores → adapters.

7.1 MainActivity.java — application entry (after manifest)

```

package com.example.miniprojectinadepartmentofficeeverydayactivitiesofaids;

import android.os.Bundle;

import androidx.activity.EdgeToEdge;

import com.google.android.material.snackbar.Snackbar;

import androidx.appcompat.app.AppCompatActivity;

import android.view.View;

import androidx.core.graphics.Insets;

```

```

import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
import androidx.navigation.NavController;
import androidx.navigation.fragment.NavHostFragment;
import androidx.navigation.ui.AppBarConfiguration;
import androidx.navigation.ui.NavigationUI;

import com.example.miniprojectinadepartmentofficeeverydayactivitiesofaids.databinding.ActivityMainBinding;

import android.view.Menu;
import android.view.MenuItem;

public class MainActivity extends AppCompatActivity {

    private AppBarConfiguration appBarConfiguration;
    private ActivityMainBinding binding;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);

        binding = ActivityMainBinding.inflate(getLayoutInflater());
        setContentView(binding.getRoot());

        ViewCompat.setOnApplyWindowInsetsListener(binding.main, (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });
        setSupportActionBar(binding.toolbar);

        NavHostFragment navHostFragment = (NavHostFragment) getSupportFragmentManager()
            .findFragmentById(R.id.nav_host_fragment_content_main);
        NavController navController = navHostFragment.getNavController();
        appBarConfiguration = new AppBarConfiguration.Builder(navController.getGraph()).build();
        NavigationUI.setupActionBarWithNavController(this, navController, appBarConfiguration);

        navController.addOnDestinationChangedListener((controller, destination, arguments) -> {
            if (destination.getId() == R.id.FirstFragment) {
                binding.fab.show();
            } else {
                binding.fab.hide();
            }
        });

        binding.fab.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                Snackbar.make(view, "Contact Department Office: ads_office@university.edu", Snackbar.LENGTH_SHORT)
                    .setAnchorView(R.id.fab)
                    .setAction("Email", null).show();
            }
        });
    }

    @Override

```

```

public boolean onCreateOptionsMenu(Menu menu) {
    getMenuInflater().inflate(R.menu.menu_main, menu);
    return true;
}

@Override
public boolean onOptionsItemSelected(MenuItem item) {
    int id = item.getItemId();

    if (id == R.id.action_settings) {
        return true;
    }

    return super.onOptionsItemSelected(item);
}

@Override
public boolean onSupportNavigateUp() {
    NavHostFragment navHostFragment = (NavHostFragment) getSupportFragmentManager()
        .findFragmentById(R.id.nav_host_fragment_content_main);
    NavController navController = navHostFragment.getNavController();
    return NavigationUI.navigateUp(navController, appBarConfiguration)
        || super.onSupportNavigateUp();
}
}

```

7.2 FirstFragment.java — dashboard navigation

```

package com.example.miniprojectinadepartmentofficeeverydayactivitiesofaids;

import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;

import androidx.annotation.NonNull;
import androidx.fragment.app.Fragment;
import androidx.navigation.fragment.NavHostFragment;

import com.example.miniprojectinadepartmentofficeeverydayactivitiesofaids.databinding.FragmentFirstBinding;

public class FirstFragment extends Fragment {

    private FragmentFirstBinding binding;

    @Override
    public View onCreateView(
        @NonNull LayoutInflater inflater, ViewGroup container,
        Bundle savedInstanceState
    ) {

        binding = FragmentFirstBinding.inflate(inflater, container, false);
        return binding.getRoot();

    }

    public void onViewCreated(@NonNull View view, Bundle savedInstanceState) {

```

```

        super.onViewCreated(view, savedInstanceState);

        binding.buttonFaculty.setOnClickListener(v ->
            NavHostFragment.findNavController(FirstFragment.this)
                .navigate(R.id.action_FirstFragment_to_SecondFragment)
        );

        binding.buttonLab.setOnClickListener(v ->
            NavHostFragment.findNavController(FirstFragment.this)
                .navigate(R.id.action_FirstFragment_to_ThirdFragment)
        );

        binding.buttonAnnouncements.setOnClickListener(v ->
            NavHostFragment.findNavController(FirstFragment.this)
                .navigate(R.id.action_FirstFragment_to_AnnouncementsFragment)
        );
    }

    @Override
    public void onDestroyView() {
        super.onDestroyView();
        binding = null;
    }
}

```

7.3 Domain models

```

// Faculty.java
package com.example.miniprojectinadepartmentofficeeverydayactivitiesofaids;

import java.io.Serializable;

public class Faculty implements Serializable {
    private String id;
    private String name;
    private String role;
    private String status;

    public Faculty(String id, String name, String role, String status) {
        this.id = id;
        this.name = name;
        this.role = role;
        this.status = status;
    }

    public String getId() { return id; }
    public String getName() { return name; }
    public void setName(String name) { this.name = name; }
    public String getRole() { return role; }
    public void setRole(String role) { this.role = role; }
    public String getStatus() { return status; }
    public void setStatus(String status) { this.status = status; }
}

// Announcement.java
package com.example.miniprojectinadepartmentofficeeverydayactivitiesofaids;

```

```

public class Announcement {
    private String id;
    private String title;
    private String description;
    private String date;

    public Announcement(String id, String title, String description, String date) {
        this.id = id;
        this.title = title;
        this.description = description;
        this.date = date;
    }

    public String getId() { return id; }
    public String getTitle() { return title; }
    public void setTitle(String title) { this.title = title; }
    public String getDescription() { return description; }
    public void setDescription(String description) { this.description = description; }
    public String getDate() { return date; }
    public void setDate(String date) { this.date = date; }
}

```

7.4 Data stores (singleton — loaded when first accessed)

// FacultyDataStore.java

```

package com.example.miniprojectinadepartmentofficeeverydayactivitiesofaids;

```

```

import java.util.ArrayList;
import java.util.List;
import java.util.UUID;

```

```

public class FacultyDataStore {
    private static FacultyDataStore instance;
    private final List<Faculty> facultyList;

    private FacultyDataStore() {
        facultyList = new ArrayList<>();
        facultyList.add(new Faculty(UUID.randomUUID().toString(), "Dr. Alice Smith", "Professor & Head", "A"));
        facultyList.add(new Faculty(UUID.randomUUID().toString(), "Dr. Bob Johnson", "Associate Professor", "B"));
    }

    public static synchronized FacultyDataStore getInstance() {
        if (instance == null) {
            instance = new FacultyDataStore();
        }
        return instance;
    }

    public List<Faculty> getFacultyList() {
        return facultyList;
    }

    public void addFaculty(Faculty faculty) {
        facultyList.add(faculty);
    }

    public void updateFaculty(Faculty updatedFaculty) {

```

```

        for (int i = 0; i < facultyList.size(); i++) {
            if (facultyList.get(i).getId().equals(updatedFaculty.getId())) {
                facultyList.set(i, updatedFaculty);
                return;
            }
        }
    }

    public void deleteFaculty(String id) {
        facultyList.removeIf(f -> f.getId().equals(id));
    }
}

// AnnouncementDataStore.java
package com.example.miniprojectinadepartmentofficeeverydayactivitiesofaids;

import java.util.ArrayList;
import java.util.List;
import java.util.UUID;

public class AnnouncementDataStore {
    private static AnnouncementDataStore instance;
    private final List<Announcement> announcementList;

    private AnnouncementDataStore() {
        announcementList = new ArrayList<>();
        announcementList.add(new Announcement(UUID.randomUUID().toString(), "Staff Meeting", "Meeting at 2
    }

    public static synchronized AnnouncementDataStore getInstance() {
        if (instance == null) {
            instance = new AnnouncementDataStore();
        }
        return instance;
    }

    public List<Announcement> getAnnouncementList() {
        return announcementList;
    }

    public void addAnnouncement(Announcement announcement) {
        announcementList.add(0, announcement); // Add to top
    }

    public void updateAnnouncement(Announcement updatedAnnouncement) {
        for (int i = 0; i < announcementList.size(); i++) {
            if (announcementList.get(i).getId().equals(updatedAnnouncement.getId())) {
                announcementList.set(i, updatedAnnouncement);
                return;
            }
        }
    }

    public void deleteAnnouncement(String id) {
        announcementList.removeIf(a -> a.getId().equals(id));
    }
}

```


7.5 SecondFragment.java — faculty UI + CRUD

```
package com.example.miniprojectinadepartmentofficeeverydayactivitiesofaids;

import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.EditText;
import android.widget.Toast;

import androidx.annotation.NonNull;
import androidx.appcompat.app.AlertDialog;
import androidx.fragment.app.Fragment;
import androidx.navigation.fragment.NavHostFragment;
import androidx.recyclerview.widget.LinearLayoutManager;

import com.example.miniprojectinadepartmentofficeeverydayactivitiesofaids.databinding.FragmentSecondBinding;

import java.util.UUID;

public class SecondFragment extends Fragment implements FacultyAdapter.OnFacultyActionListener {

    private FragmentSecondBinding binding;
    private FacultyAdapter adapter;

    @Override
    public View onCreateView(
        @NonNull LayoutInflater inflater, ViewGroup container,
        Bundle savedInstanceState
    ) {
        binding = FragmentSecondBinding.inflate(inflater, container, false);
        return binding.getRoot();
    }

    public void onViewCreated(@NonNull View view, Bundle savedInstanceState) {
        super.onViewCreated(view, savedInstanceState);

        setupRecyclerView();

        binding.buttonSecond.setOnClickListener(v ->
            NavHostFragment.findNavController(SecondFragment.this)
                .navigate(R.id.action_SecondFragment_to_FirstFragment)
        );

        binding.fabAddFaculty.setOnClickListener(v -> showFacultyDialog(null));
    }

    private void setupRecyclerView() {
        adapter = new FacultyAdapter(FacultyDataStore.getInstance().getFacultyList(), this);
        binding.recyclerViewFaculty.setLayoutManager(new LinearLayoutManager(getContext()));
        binding.recyclerViewFaculty.setAdapter(adapter);
        updateEmptyView();
    }

    private void updateEmptyView() {
        if (FacultyDataStore.getInstance().getFacultyList().isEmpty()) {
```

```

        binding.textNoFaculty.setVisibility(View.VISIBLE);
        binding.recyclerViewFaculty.setVisibility(View.GONE);
    } else {
        binding.textNoFaculty.setVisibility(View.GONE);
        binding.recyclerViewFaculty.setVisibility(View.VISIBLE);
    }
}

private void showFacultyDialog(Faculty faculty) {
    View dialogView = LayoutInflater.from(getContext()).inflate(R.layout.dialog_faculty, null);
    EditText nameEdit = dialogView.findViewById(R.id.edit_faculty_name);
    EditText roleEdit = dialogView.findViewById(R.id.edit_faculty_role);
    EditText statusEdit = dialogView.findViewById(R.id.edit_faculty_status);

    if (faculty != null) {
        nameEdit.setText(faculty.getName());
        roleEdit.setText(faculty.getRole());
        statusEdit.setText(faculty.getStatus());
    }

    new AlertDialog.Builder(requireContext())
        .setTitle(faculty == null ? R.string.add_faculty_title : R.string.edit_faculty_title)
        .setView(dialogView)
        .setPositiveButton(R.string.confirm_button, (dialog, which) -> {
            String name = nameEdit.getText().toString();
            String role = roleEdit.getText().toString();
            String status = statusEdit.getText().toString();

            if (faculty == null) {
                Faculty newFaculty = new Faculty(UUID.randomUUID().toString(), name, role, status);
                FacultyDataStore.getInstance().addFaculty(newFaculty);
                Toast.makeText(getContext(), "Faculty added", Toast.LENGTH_SHORT).show();
            } else {
                faculty.setName(name);
                faculty.setRole(role);
                faculty.setStatus(status);
                FacultyDataStore.getInstance().updateFaculty(faculty);
                Toast.makeText(getContext(), "Faculty updated", Toast.LENGTH_SHORT).show();
            }
            adapter.notifyDataSetChanged();
            updateEmptyView();
        })
        .setNegativeButton(R.string.cancel_button, null)
        .show();
}

@Override
public void onEdit(Faculty faculty) {
    showFacultyDialog(faculty);
}

@Override
public void onDelete(Faculty faculty) {
    new AlertDialog.Builder(requireContext())
        .setTitle(R.string.delete_faculty_title)
        .setMessage(getString(R.string.delete_faculty_message, faculty.getName()))
        .setPositiveButton(R.string.delete_button, (dialog, which) -> {

```

```

        FacultyDataStore.getInstance().deleteFaculty(faculty.getId());
        Toast.makeText(getContext(), "Faculty deleted", Toast.LENGTH_SHORT).show();
        adapter.notifyDataSetChanged();
        updateEmptyView();
    })
    .setNegativeButton(R.string.cancel_button, null)
    .show();
}

@Override
public void onDestroyView() {
    super.onDestroyView();
    binding = null;
}
}

```

7.6 ThirdFragment.java — lab booking

```

package com.example.miniprojectinadepartmentofficeeverydayactivitiesofaids;

import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.Toast;

import androidx.annotation.NonNull;
import androidx.fragment.app.Fragment;
import androidx.navigation.fragment.NavHostFragment;

import com.example.miniprojectinadepartmentofficeeverydayactivitiesofaids.databinding.FragmentThirdBinding;

public class ThirdFragment extends Fragment {

    private FragmentThirdBinding binding;

    @Override
    public View onCreateView(
        @NonNull LayoutInflater inflater, ViewGroup container,
        Bundle savedInstanceState
    ) {

        binding = FragmentThirdBinding.inflate(inflater, container, false);
        return binding.getRoot();

    }

    public void onViewCreated(@NonNull View view, Bundle savedInstanceState) {
        super.onViewCreated(view, savedInstanceState);

        binding.buttonSubmitBooking.setOnClickListener(v -> {
            String lab = binding.editLabName.getText().toString();
            if (!lab.isEmpty()) {
                Toast.makeText(getContext(), "Booking requested for " + lab, Toast.LENGTH_SHORT).show();
            } else {
                Toast.makeText(getContext(), "Please enter a lab name", Toast.LENGTH_SHORT).show();
            }
        })
    }
}

```

```

    });

    binding.buttonThirdBack.setOnClickListener(v ->
        NavHostFragment.findNavController(ThirdFragment.this)
            .navigate(R.id.action_ThirdFragment_to_FirstFragment)
    );
}

@Override
public void onDestroyView() {
    super.onDestroyView();
    binding = null;
}
}

```

7.7 AnnouncementsFragment.java

```

package com.example.miniprojectinadepartmentofficeeverydayactivitiesofaids;

import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.EditText;
import android.widget.Toast;

import androidx.annotation.NonNull;
import androidx.appcompat.app.AlertDialog;
import androidx.fragment.app.Fragment;
import androidx.navigation.fragment.NavHostFragment;
import androidx.recyclerview.widget.LinearLayoutManager;

import com.example.miniprojectinadepartmentofficeeverydayactivitiesofaids.databinding.FragmentAnnouncements;

import java.text.SimpleDateFormat;
import java.util.Date;
import java.util.Locale;
import java.util.UUID;

public class AnnouncementsFragment extends Fragment implements AnnouncementAdapter.OnAnnouncementActionListener {

    private FragmentAnnouncementsBinding binding;
    private AnnouncementAdapter adapter;

    @Override
    public View onCreateView(
        @NonNull LayoutInflater inflater, ViewGroup container,
        Bundle savedInstanceState
    ) {
        binding = FragmentAnnouncementsBinding.inflate(inflater, container, false);
        return binding.getRoot();
    }

    public void onViewCreated(@NonNull View view, Bundle savedInstanceState) {
        super.onViewCreated(view, savedInstanceState);
    }
}

```

```

        adapter = new AnnouncementAdapter(AnnouncementDataStore.getInstance().getAnnouncementList(), this);
        binding.recyclerViewAnnouncements.setLayoutManager(new LinearLayoutManager(getContext()));
        binding.recyclerViewAnnouncements.setAdapter(adapter);
        updateEmptyView();

        binding.buttonAnnouncementsBack.setOnClickListener(v ->
            NavHostFragment.findNavController(AnnouncementsFragment.this)
                .navigate(R.id.action_AnnouncementsFragment_to_FirstFragment)
        );

        binding.fabAddAnnouncement.setOnClickListener(v -> showAnnouncementDialog(null));
    }

    private void updateEmptyView() {
        if (AnnouncementDataStore.getInstance().getAnnouncementList().isEmpty()) {
            binding.textNoAnnouncements.setVisibility(View.VISIBLE);
            binding.recyclerViewAnnouncements.setVisibility(View.GONE);
        } else {
            binding.textNoAnnouncements.setVisibility(View.GONE);
            binding.recyclerViewAnnouncements.setVisibility(View.VISIBLE);
        }
    }

    private void showAnnouncementDialog(Announcement announcement) {
        View dialogView = LayoutInflater.from(getContext()).inflate(R.layout.dialog_announcement, null);
        EditText titleEdit = dialogView.findViewById(R.id.edit_announcement_title);
        EditText descEdit = dialogView.findViewById(R.id.edit_announcement_desc);

        if (announcement != null) {
            titleEdit.setText(announcement.getTitle());
            descEdit.setText(announcement.getDescription());
        }

        new AlertDialog.Builder(requireContext())
            .setTitle(announcement == null ? R.string.add_announcement_title : R.string.edit_announcement_title)
            .setView(dialogView)
            .setPositiveButton(R.string.post_button, (dialog, which) -> {
                String title = titleEdit.getText().toString();
                String desc = descEdit.getText().toString();

                if (announcement == null) {
                    String date = new SimpleDateFormat("MMM dd, yyyy", Locale.getDefault()).format(new Date());
                    Announcement newAnnouncement = new Announcement(UUID.randomUUID().toString(), title, desc, date);
                    AnnouncementDataStore.getInstance().addAnnouncement(newAnnouncement);
                    Toast.makeText(getContext(), "Announcement posted", Toast.LENGTH_SHORT).show();
                } else {
                    announcement.setTitle(title);
                    announcement.setDescription(desc);
                    AnnouncementDataStore.getInstance().updateAnnouncement(announcement);
                    Toast.makeText(getContext(), "Announcement updated", Toast.LENGTH_SHORT).show();
                }
                adapter.notifyDataSetChanged();
                updateEmptyView();
            })
            .setNegativeButton(R.string.cancel_button, null)
            .show();
    }
}

```

```

@Override
public void onEdit(Announcement announcement) {
    showAnnouncementDialog(announcement);
}

@Override
public void onDelete(Announcement announcement) {
    new AlertDialog.Builder(requireContext())
        .setTitle(R.string.delete_announcement_title)
        .setMessage(R.string.delete_announcement_message)
        .setPositiveButton(R.string.delete_button, (dialog, which) -> {
            AnnouncementDataStore.getInstance().deleteAnnouncement(announcement.getId());
            Toast.makeText(getContext(), "Announcement deleted", Toast.LENGTH_SHORT).show();
            adapter.notifyDataSetChanged();
            updateEmptyView();
        })
        .setNegativeButton(R.string.cancel_button, null)
        .show();
}

@Override
public void onDestroyView() {
    super.onDestroyView();
    binding = null;
}
}

```

7.8 RecyclerView adapters (bind row XML → model fields)

```

// FacultyAdapter.java - full file
package com.example.miniprojectinadepartmentofficeeverydayactivitiesofaids;

import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.ImageButton;
import android.widget.TextView;

import androidx.annotation.NonNull;
import androidx.recyclerview.widget.RecyclerView;

import java.util.List;

public class FacultyAdapter extends RecyclerView.Adapter<FacultyAdapter.FacultyViewHolder> {

    private final List<Faculty> facultyList;
    private final OnFacultyActionListener listener;

    public interface OnFacultyActionListener {
        void onEdit(Faculty faculty);
        void onDelete(Faculty faculty);
    }

    public FacultyAdapter(List<Faculty> facultyList, OnFacultyActionListener listener) {
        this.facultyList = facultyList;
        this.listener = listener;
    }
}

```

```

    }

    @NonNull
    @Override
    public FacultyViewHolder onCreateViewHolder(@NonNull ViewGroup parent, int viewType) {
        View view = LayoutInflater.from(parent.getContext()).inflate(R.layout.item_faculty, parent, false);
        return new FacultyViewHolder(view);
    }

    @Override
    public void onBindViewHolder(@NonNull FacultyViewHolder holder, int position) {
        Faculty faculty = facultyList.get(position);
        holder.nameText.setText(faculty.getName());
        holder.roleText.setText(faculty.getRole());
        holder.statusText.setText(faculty.getStatus());

        holder.editButton.setOnClickListener(v -> listener.onEdit(faculty));
        holder.deleteButton.setOnClickListener(v -> listener.onDelete(faculty));
    }

    @Override
    public int getItemCount() {
        return facultyList.size();
    }

    static class FacultyViewHolder extends RecyclerView.ViewHolder {
        TextView nameText, roleText, statusText;
        ImageButton editButton, deleteButton;

        public FacultyViewHolder(@NonNull View itemView) {
            super(itemView);
            nameText = itemView.findViewById(R.id.text_faculty_name);
            roleText = itemView.findViewById(R.id.text_faculty_role);
            statusText = itemView.findViewById(R.id.text_faculty_status);
            editButton = itemView.findViewById(R.id.button_edit_faculty);
            deleteButton = itemView.findViewById(R.id.button_delete_faculty);
        }
    }
}

// AnnouncementAdapter.java - full file
package com.example.miniprojectinadepartmentofficeeverydayactivitiesofaids;

import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.ImageButton;
import android.widget.TextView;

import androidx.annotation.NonNull;
import androidx.recyclerview.widget.RecyclerView;

import java.util.List;

public class AnnouncementAdapter extends RecyclerView.Adapter<AnnouncementAdapter.AnnouncementViewHolder> {

    private final List<Announcement> announcementList;

```

```

private final OnAnnouncementActionListener listener;

public interface OnAnnouncementActionListener {
    void onEdit(Announcement announcement);
    void onDelete(Announcement announcement);
}

public AnnouncementAdapter(List<Announcement> announcementList, OnAnnouncementActionListener listener) {
    this.announcementList = announcementList;
    this.listener = listener;
}

@NonNull
@Override
public AnnouncementViewHolder onCreateViewHolder(@NonNull ViewGroup parent, int viewType) {
    View view = LayoutInflater.from(parent.getContext()).inflate(R.layout.item_announcement, parent, false);
    return new AnnouncementViewHolder(view);
}

@Override
public void onBindViewHolder(@NonNull AnnouncementViewHolder holder, int position) {
    Announcement announcement = announcementList.get(position);
    holder.titleText.setText(announcement.getTitle());
    holder.descriptionText.setText(announcement.getDescription());
    holder.dateText.setText(announcement.getDate());

    holder.editButton.setOnClickListener(v -> listener.onEdit(announcement));
    holder.deleteButton.setOnClickListener(v -> listener.onDelete(announcement));
}

@Override
public int getItemCount() {
    return announcementList.size();
}

static class AnnouncementViewHolder extends RecyclerView.ViewHolder {
    TextView titleText, descriptionText, dateText;
    ImageButton editButton, deleteButton;

    public AnnouncementViewHolder(@NonNull View itemView) {
        super(itemView);
        titleText = itemView.findViewById(R.id.text_announcement_title);
        descriptionText = itemView.findViewById(R.id.text_announcement_description);
        dateText = itemView.findViewById(R.id.text_announcement_date);
        editButton = itemView.findViewById(R.id.button_edit_announcement);
        deleteButton = itemView.findViewById(R.id.button_delete_announcement);
    }
}
}

```

7.9 View Binding name mapping (study aid)

Layout file	Generated binding class	Example field
activity_main.xml	ActivityMainBinding	binding.toolbar, binding.fab
fragment_first.xml	FragmentFirstBinding	binding.buttonFaculty

Layout file	Generated binding class	Example field
fragment_second.xml	FragmentSecondBinding	binding.recyclerViewFaculty
item_faculty.xml	(manual inflate in adapter)	R.id.text_faculty_name

8. Complete Lifecycle Flow (Summary)

8.1 Activity lifecycle (MainActivity)

Callback	What happens in this project
onCreate	Inflate UI, NavController, FAB rules, menu
onCreateOptionsMenu	Inflate Settings menu
onOptionsItemSelected	Settings stub (no action yet)
onSupportNavigateUp	Pop fragment back stack via Navigation

8.2 Fragment lifecycle (typical screen)

```
onAttach → onCreate → onCreateView (inflate XML)
  → onViewCreated (listeners, RecyclerView)
  → onStart → onResume [user interacts]
  → onPause → onStop
  → onDestroyView (binding = null)
  → onDestroy → detach
```

8.3 RecyclerView item lifecycle

```
Adapter.onCreateViewHolder → inflates item_*.xml
Adapter.onBindViewHolder → sets TextViews from model
User taps Edit/Delete → listener → Fragment dialog / confirm
notifyDataSetChanged() → rebinds visible rows
```

8.4 End-to-end “Add faculty” timeline

1. XML fragment_second.xml defines fab_add_faculty.
2. Java SecondFragment.onViewCreated sets click listener.
3. Java showFacultyDialog inflates dialog_faculty.xml.
4. User taps Confirm → FacultyDataStore.addFaculty.
5. Java adapter.notifyDataSetChanged() → onBindViewHolder updates list.

9. File Relationship Diagram

9.1 Mermaid — component dependencies

```
graph TD
    subgraph config [Configuration]
        M[AndroidManifest.xml]
        NG[nav_graph.xml]
        BG[app/build.gradle.kts]
    end

    subgraph activity [Activity Layer]
        MA[MainActivity.java]
        AM[activity_main.xml]
        CM[content_main.xml]
    end
```

```

end

subgraph fragments [Fragments]
    F1[FirstFragment]
    F2[SecondFragment]
    F3[ThirdFragment]
    FA[AnnouncementsFragment]
end

subgraph data [Data Layer]
    FD[FacultyDataStore]
    AD[AnnouncementDataStore]
    FM[Faculty.java]
    AN[Announcement.java]
end

subgraph ui_lists [Lists]
    FAd[FacultyAdapter]
    AAd[AnnouncementAdapter]
    IF[item_faculty.xml]
    IA[item_announcement.xml]
end

M --> MA
NG --> CM
AM --> MA
CM --> MA
MA --> F1
F1 --> F2
F1 --> F3
F1 --> FA
F2 --> FD
F2 --> FAd
FAd --> FM
FAd --> IF
FD --> FM
FA --> AD
FA --> AAd
AAd --> AN
AAd --> IA
AD --> AN
BG --> MA

```

9.2 ID reference table (XML Java)

XML @+id	Java reference
nav_host_fragment_content_main	MainActivity, NavHostFragment
button_faculty	FragmentFirstBinding.buttonFaculty
recycler_view_faculty	FragmentSecondBinding.recyclerViewFaculty
edit_faculty_name	dialogView.findViewById in SecondFragment
edit_lab_name	FragmentThirdBinding.editLabName
recycler_view_announcements	FragmentAnnouncementsBinding.recyclerViewAnnouncements

10. Lab Exercises for Students

Exercise 1 — Trace the launch path

Draw a sequence diagram from app icon tap to `FirstFragment` visible. Label each file involved.

Exercise 2 — String resources

Add a new string `department_tagline` in `strings.xml` and display it under the welcome `TextView` in `fragment_first.xml`.

Exercise 3 — Navigation

Add a fourth button on the dashboard labeled **About Department** that navigates to a new `AboutFragment` (create layout, fragment class, and `nav_graph` entry).

Exercise 4 — Validation

In `SecondFragment.showFacultyDialog`, show a `Toast` and **do not save** if name or role is empty.

Exercise 5 — Lab booking enhancement

Update `ThirdFragment` so the `Toast` includes date and time from `edit_date` and `edit_time` when provided.

Exercise 6 — RecyclerView

Change `FacultyAdapter` to use **View Binding** for `item_faculty.xml` instead of `findViewById`.

Exercise 7 — Persistence (advanced)

Replace `FacultyDataStore` in-memory list with `SharedPreferences` or `Room` so data survives app restart.

Exercise 8 — Settings menu

Implement `MainActivity.onOptionsItemSelected` for Settings: show an `AlertDialog` with app version from `BuildConfig.VERSION_NAME`.

Submission checklist

- ☐ Screenshots of all four main screens
 - ☐ Answers to Exercise 1 sequence diagram
 - ☐ Git commit history or zip with modified source
 - ☐ Short report (1 page): fragments vs activities, why singleton datastore was used
-

11. Prerequisites Checklist

11.1 Software

Item	Minimum	Verify
Android Studio	Latest stable	Help → About
JDK	11+	File → Project Structure → SDK
Android SDK	API 26–36	SDK Manager
Emulator or device	API 26+	Device Manager

11.2 Knowledge

- ☐ Java classes, interfaces, `ArrayList`
- ☐ XML layouts and `@string/` references
- ☐ Activity vs Fragment lifecycle basics
- ☐ Intents and launcher activity concept
- ☐ Optional: RecyclerView pattern (`ViewHolder`, `Adapter`)

11.3 Opening the project

1. **File** → **Open** → select project root folder.
 2. Wait for Gradle sync to finish.
 3. **Run** on **app** module.
 4. Confirm dashboard shows: *Welcome to AI & DS Department Office*.
-

12. Troubleshooting

Problem	Likely cause	Solution
Gradle sync failed	Network / AGP version	Check internet; File → Sync Project with Gradle Files
R.id cannot be resolved	XML syntax error or wrong import	Fix red underlines in <code>res/</code> ; Build → Clean Project → Rebuild
App crashes on launch	Nav graph / fragment class mismatch	Ensure <code>android:name</code> in <code>nav_graph.xml</code> matches full package class name
Blank RecyclerView	Empty list or adapter not set	Add sample data in <code>DataStore</code> ; call <code>setAdapter</code> and <code>setLayoutManager</code>
FAB not visible	Not on <code>FirstFragment</code>	By design: FAB shows only on Dashboard (<code>MainActivity</code> listener)
View Binding errors	<code>viewBinding</code> false or stale build	Confirm <code>viewBinding = true</code> in <code>app/build.gradle.kts</code> ; rebuild
<code>onViewCreated</code> never runs	Using wrong callback registration	Fragments from Navigation call <code>onViewCreated</code> automatically when using default constructor
Data lost after kill app	In-memory store only	Expected behavior; complete Exercise 7 for persistence
Up/back does not work	<code>AppBarConfiguration</code> not set	Verify <code>NavigationUI.setupActionBarWithNavController</code> in <code>MainActivity</code>
Emulator too slow	Low RAM allocation	Enable hardware acceleration; use physical device if possible

Logcat tips

Filter by package:

```
package:com.example.miniprojectinadepartmentofficeeverydayactivitiesofaids
```

Look for **FATAL EXCEPTION** and read the **Caused by** line — it usually names the missing ID, null context, or navigation action.

Appendix A — Quick reference: execution order cheat sheet

1. `AndroidManifest.xml`
 2. `themes.xml` (application theme)
 3. `strings.xml` / `dimens.xml` (resources)
 4. `MainActivity.onCreate` → `activity_main.xml` + `content_main.xml`
 5. `nav_graph.xml` → `FirstFragment`
 6. `fragment_first.xml` + `FirstFragment.java`
 7. [User action] → target fragment layout + Java
 8. [List screens] `DataStore` → `Adapter` → `item_*.xml`
 9. [Dialogs] `dialog_*.xml` inflated inside `AlertDialog.Builder`
-

Appendix B — Distribution notes for instructors

- Export this file to PDF via Android Studio, Pandoc, or any Markdown-to-PDF tool for print-ready handouts.
 - Pair with a live demo: add faculty → rotate device → discuss why data may or may not survive (configuration changes vs process death).
 - Encourage students to use **Layout Inspector** and **Navigation Editor** while following Sections 3 and 9.
-

Document generated from project source at `app/src/main/`. Align lab dates and course codes on the cover page when distributing.